Guided Step-by-Step Demo: Configuring Autoscaling for Managed Instance Groups in Google Cloud

Overview & Objectives

In this hands-on lab demo, you will learn how to configure an autoscaling policy on a Managed Instance Group (MIG) in Google Cloud. The MIG will automatically scale out (add VMs) when CPU utilization exceeds a specified threshold and scale in (remove VMs) when demand drops.

By the end of this demo, you will be able to:

1. Create a template and regional Managed Instance Group.
2. Attach an autoscaling policy based on CPU utilization and initialization periods.
3. Simulate load to trigger autoscaling events and verify automatic scaling behavior.

Task 1: Create an Instance Template

1. Navigate to **Compute Engine > Instance templates** and click **Create instance template**.
    - **Name:** `autoscale-template`
    - **Machine type:** `e2-micro`
    - **Boot disk:** Ubuntu 24.04 LTS
    - Expand **Management, security, disk, networking, sole tenancy** -> **Management** -> **Startup script**:

      #!/bin/bash

      apt-get update

      apt-get install -y stress

    - Click **Create**.

Task 2: Create a Regional Managed Instance Group (MIG)

1. Navigate to **Compute Engine > Instance groups** and click **Create instance group**:

- **Name:** `regional-autoscale-mig`
- **Instance group type:** Managed instance group (stateless)
- **Instance template:** `autoscale-template`
- **Location:** Regional (select regions such as `us-central1` across multiple zones for resilience).
- **Autoscaling:** Click *Configure autoscaling*.
    - **Autoscaling metric:** CPU utilization
    - **Target CPU utilization:** `60%`
    - **Min number of instances:** `2`
    - **Max number of instances:** `6`
    - **Cool down period:** `60 seconds`
- **Initial delay (Autohealing):** Set to `120 seconds` (ensures applications finish booting before health checks judge them).
- Click **Create**.

Task 3: Trigger Autoscaling via CPU Stress Test

1. Go to **Compute Engine > VM instances**. You will see 2 running instances created by your regional MIG.
2. Click **SSH** on one of the active instances.
3. Install and run a CPU stress tool to push CPU usage above 60%:

   sudo apt-get install -y stress

   stress --cpu 2 --timeout 300

4. Return to **Compute Engine > Instance groups**, click on `regional-autoscale-mig`, and monitor the **Monitoring** tab.
5. **Observation:** Within 2–3 minutes, notice how the instance count automatically scales up from 2 toward your maximum limit to handle the load. Once the stress command terminates, the MIG will automatically scale back down.